SaaS Hiring & Resume Intelligence Platform (ATS + AI)

Project Overview

This project is a full-stack SaaS-based Applicant Tracking System (ATS) built using the MERN stack with AI-powered resume analysis. It helps recruiters efficiently screen candidates, candidates apply for jobs, and admins monitor platform analytics.

Core Objectives

- Build a production-grade MERN application
- Implement role-based access control
- Integrate AI for resume-job matching
- Design analytics dashboards with charts
- Follow real-world SaaS architecture

User Roles

1. Candidate: Apply for jobs and upload resumes
2. Recruiter: Post jobs, view applicants, and shortlist candidates
3. Admin: Monitor platform analytics and manage users

Tech Stack

Frontend: React.js, Redux Toolkit, Tailwind CSS, Recharts
Backend: Node.js, Express.js, JWT, RBAC
Database: MongoDB, Mongoose
AI: Google Gemini API (Resume Parsing & Matching)
Cloud & Deployment: Vercel, Render, MongoDB Atlas, Cloudinary

System Architecture

The frontend communicates with REST APIs built in Express. Authentication is handled using JWT tokens. Resumes are uploaded to cloud storage. AI services analyze resumes and return match scores, which are stored in MongoDB.

AI Integration Workflow

1. Resume uploaded by candidate
2. PDF converted to text
3. AI compares resume with job description

4. Match score and skill gap generated
5. Results shown on recruiter dashboard

Dashboards & Analytics

Recruiter Dashboard: Applications per job, AI score distribution
Admin Dashboard: User growth, jobs per month
Candidate Dashboard: Application status and resume feedback

Implementation Phases

Phase 1: Backend APIs & Authentication
Phase 2: Frontend UI & Dashboards
Phase 3: AI Integration
Phase 4: Deployment & Optimization

Outcome

This project demonstrates real-world system design, backend engineering, AI integration, and SaaS product thinking, making it ideal for interviews and resume shortlisting.